Formal Architecture Guide for HLS Scheduling Rules

Constructive safety and required workload completion

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This guide explains how the current scheduling rules support source-level verification of an HLS implementation. The architecture has two principal results: constructive correspondence transfers observable safety from the configured source model to the RTL; required-workload completion adds a finite accepting source witness and implementation progress assumptions to prove eventual acceptance of the required outputs. The two results share semantics and evidence, but establish different guarantees.

This is an informative companion to the Catapult User View Scheduling Rules document, available here:

https://github.com/Stuart-Swan/Matchlib-Examples-Kit-For-Accellera-Synthesis-WG/blob/master/matchlib_examples/doc/catapult_user_view_scheduling_rules.pdf

That document remains authoritative for definitions, theorem premises and applicability. Named references below identify its controlling interfaces. This edition replaces the September guide's replay, certificate and strict-deadlock architecture with the retained constructive and completion routes. It is organized for architects, verification engineers and formal-methods implementers who need to understand what must be proved and where evidence can be reused.

1 The architecture and its guarantees

The source reference is not automatically the original SystemC simulation. It is Sys_ref, configured with the implementation's relevant storage, channel boundaries, coupling and environment. The implementation is RTL_B. A theorem instance fixes these models, their initial and reset semantics, stimulus and choices, observation rules, and evidence bindings before correspondence is attempted.

For safety, Appendix J constructs a legal source execution prefix for every covered finite implementation prefix. The two prefixes have equal canonical observable histories. If the required property holds on every history in the independently declared source domain, it therefore holds on the covered implementation domain. This is the normal implementation-to-source direction: every covered RTL behavior is explained by a legal source behavior. It does not require the RTL to reproduce an arbitrarily selected source simulation.

For completion, Appendix K fixes a finite collection R of required testbench acceptances. A qualified finite source execution accepts R. Internal-fact correspondence and conditional agreement connect that witness to the RTL; controller, resource and service laws ensure the required operations actually occur. K.CT.Completion then establishes eventual acceptance of R with the witness's values and per-endpoint order for every qualified implementation execution satisfying those service assumptions.

Result	Main inputs	What follows

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Constructive history coverage	Local source and implementation proofs, joint settling, environment, order and actual prefix carrier	Every covered RTL prefix has an equal reachable source history
Source safety transfer	History coverage and an independent property proof on the selected source domain	The extensional history property holds on the covered RTL domain
Internal-fact correspondence	Constructive local fact laws and exact coverage of the consumed fact signature	Completed internal and external facts match legal source facts
Required-workload completion	Full facts, agreement, finite witness, exposure and fair service	Required testbench acceptances eventually occur with matching values and endpoint order

Safety does not imply eventual output. Completion does not imply a cycle bound, a total order across unrelated endpoints, exclusion of extra outputs, or absence of every internal deadlock. Extra-output exclusion is a safety property. A deadlock after R has been accepted, or in unrelated activity, is outside K's conclusion. Source and implementation need not have equal registers, identical cycle counts or the same grouping of independent events into cycles.

The September architecture is no longer the application path. Appendix I is reserved. The former J.1/GWR certificate stack, KObs and wait-for-graph/Policy-S dominance argument are not the retained proof chain. M.7b, M.7c and M.8 are retired; there is no single-RTL-run B1 safety shortcut. B1 remains a separately claimable observable determinism property, not a substitute for constructive coverage, source safety or progress. Appendices R and S now describe the retained dependencies and mechanization targets.

2 Models, boundaries and observable histories

Selecting the source reference

Core.1 selects Sys_B for ordinary abstract channels. B(c)=0 means a primitive rendezvous: the matching Push and Pop commit together with the same value. B(c)>0 means a bounded, cycle-boundary, non-fall-through FIFO. A Pop needs an old item; a Push needs old free space. An initially full primitive FIFO cannot accept a Push merely because a Pop occurs on the same edge, and an initially empty one cannot satisfy a Pop with that edge's new Push.

Sys_B^elab(E) is used when a scalar depth does not faithfully describe the communication structure. It explicitly represents finite storage and the stateless deterministic couplers needed for staging, joins, bundles or synchronization gates. A nonempty or nonfull storage element may still be unavailable at a coupled process boundary: actual enablement uses the complete settled handshake equations. Storage or credit hidden from the model can change both safety and completion applicability.

Q.SafetyQualified(E) binds the elaborated state, accounting, ownership, effects, atomic groups, required order, frontier attribution and handshake network. It does not itself prove source/RTL correspondence, whole-system settling or progress. Q.CTQualified adds the applicable completion laws. Mathematical admissibility is also

distinct from flow availability: a qualified model for which no executable or independently checkable source artifact is available cannot yet support the corresponding executable evidence workflow.

Keeping accounting and observation separate

The proof-internal alphabet Σ includes committed transfers at explicit modeled boundaries, selected synchronization and signal observations. An introduced staging transfer may be hidden from the DUT interface and still be a proof-internal event. Failed NB attempts emit no message event, but their outcomes and identities may matter to later behavior or to the completion fact signature.

For elaboration, $\pi_{\Sigma,E}$ produces the complete outer accounting history, including both Push and Pop at every outer channel. Only afterward does ρ_{DUT} restrict that history to Σ_{DUT}. Thus $\pi_{DUT,E} = \rho_{DUT} \circ \pi_{\Sigma,E}$. FIFO balance is checked before DUT observation restriction; hiding a consumer event does not remove its accounting effect.

The state-computed signed map A_E must equal the channel-epoch Push-minus-Pop balance at the specified cycle-entry boundary. Its legal nonnegative range follows from that agreement and outer E4 legality, not from its signed type. Shared accounting views do not create extra storage or permit the same credit to be spent twice. Q.1 requires explicit ownership and invariant preservation for shared support.

J.OuterCanonHist records complete observable groups, occurrence identities, values and the required partial order. A rendezvous pair, paired synchronization, or qualified multi-effect atom is indivisible. Projection retains a complete selected group or erases it; it cannot retain an unauthorized half. Required paths through erased units survive as order between retained groups. Erasing a synchronization or reset therefore need not erase its ordering effect.

Three kinds of order

Source issue order, canonical observable order and proof-construction order serve different purposes. E3 constrains dynamic source-call Issue order and prefix closure, including across overlapped iterations. It does not automatically add a strict committed-event edge between distinct interfaces. The canonical order contains only the normative channel, synchronization, signal, reset and selected extension constraints. A descriptor ordering used to execute the proof is a legal witness serialization, not an additional observable dependency.

J.GeneratorCompleteness inventories every applicable ordering rule. J.RequiredOrderWF requires predecessor closure, stability of order between existing occurrences, and joint acyclicity. These are separate obligations: individually acyclic rule families can form a cycle when combined. Source primitive realization and a common rank can establish strictness; J.SourceOrderReflection can then transfer it to retained RTL groups before canonical equality is concluded. Any consumed erased-only graph still needs its own joint-acyclicity evidence under J.RequiredOrderWF.

A finite ideal of a canonical history retains whole groups and every required predecessor of each retained group. It need not be a prefix of one logger ordering, and it need not be operationally reachable. This distinction explains why selected-run property proofs and the alternative cycle-ideal route have additional conditions.

3 The operational contract behind the proof

Scheduling and request attribution

Appendix G's R1–R5 state source synchronization order, signal anchoring, isolation around synchronization, message issue order and channel semantics. E1–E5 preserve the corresponding implementation obligations,

together with Core.17's occurrence and value matching. The compatibility predicate is ordered: its left argument is the source witness and its right argument is the implementation. It is not an unrestricted mathematical equivalence relation.

Issue is the first correctly attributed process-owned request assertion for the dynamic call. Commit is its selected process-facing completion. A raw ready/valid level or transfer pulse gated by complementary availability is not automatically the process request. Core.10 and Core.IC require a fixed, bidirectionally complete mapping from requests to their actual owners. An issued blocking request and stable Push payload persist until completion or an affecting reset; back-to-back requests retain exact next-owner attribution.

For distinct blocking calls on the same scalar explicit abstract boundary and endpoint direction within one reset epoch, Core.IC(4) and endpoint serialization give strictly increasing Issue cycles. Blocking Pops on different channels may issue in the same cycle: E3's distinct-interface constraint is non-strict. Distinct-interface no-reverse order and cross-interface issue-prefix closure still require their own evidence. A physical early read or movement inside a pipeline cannot be relabeled as a different source Issue merely to satisfy E3.

E5 also includes the completion fence: blocking calls in the preceding source interval must process-facing commit no later than their synchronization. Same-edge message and synchronization completion is permitted, but the canonical message-before-sync predecessor remains mandatory. New following-interval Issue is strictly later. A physical sub-cycle timestamp is not required to express that semantic precedence.

Preparation, settling and registered results

Core.Phase gives the source operational interpretation. In the L1 preparation window, a finite selection of legal evaluation and Issue steps may occur using already available operands. L2 freezes the relevant roots and settles the complete network. L3 realizes the selected boundary actions from that snapshot. L4 installs registered results and updates; evaluation or Issue depending on a cycle-t return begins only at a strictly later Horizon under Core.RegSeq.

Core.SelectLatitude imposes no maximal-progress requirement on the selected finite L1 issue window or L3 boundary-action set. An otherwise legal action may remain unselected; nonselection does not mean it was disabled. This latitude remains subject to the actual NB decision law and all phase, source-order and Core.16 restrictions. Complete-execution fairness, when consumed, is a separate obligation.

The construction must prove those source steps legal under Core.ReachPrep, Core.SourceChoiceBinding and Core.PreparationEvidence. Constructor success is not the definition of legal preparation. An NB result follows Core.NBDecision from the actual qualified snapshot and fixed resolutions; it cannot be chosen by an annotation or manufactured by declining a transfer that the decision law requires.

J.SourceSettleRealization supplies a finite legal path to the first operational L2 fixed point and matching boundary values. The whole combined graph must include all relevant requests and absences, nonminimal pending calls, NB attempts, environment logic, guards and frontier predicates. Qualifying each coupler separately is insufficient when cross-component dependencies have not been checked.

Ordinary locality offers two source-settling paths: QF reuses a qualified source QSync result; QSafe supplies the narrower safety-only operational evidence. Both retain finite deterministic acyclic evaluation and require QM source/implementation snapshot matching. Separate QSync qualifications do not themselves prove matching. QSync also does not prove boundary-update commutation, process refinement or Batch prefix semantics. The retained sufficient constructive profiles do not acquire combinational-feedback support merely because a broader fixed-point argument might be conceivable.

Blocking does not freeze all physical computation

Core.16 activates at the first settled snapshot where the nonempty pending blocking frontier is fully disabled. It does not retroactively invalidate legal eager issues from the preceding L1 window. While that activation persists, it forbids new source-later launch, including source evaluation as well as Issue, but permits qualified progress of already-admitted work.

Physical private computation may continue if it is abstractly invisible: it changes neither represented source progress nor protected requests, payloads, storage, readiness or admitted work. Later use must match the legally evaluated source operands and their writer/occurrence identities, respect registered-return timing, and remain legal under any persisting activation. Computing privately from a FIFO head does not authorize a source computation or dependent request before the corresponding Pop's legal return use.

Core.16 and its finite decreasing drain measure constrain permitted behavior. They do not force movement. B.OF and B.CT separately supply the positive implementation laws needed for completion: eventual request assertion, finite serviced computation, ordered movement and useful downstream drainage.

4 How constructive correspondence is established

The local construction

J.ConstructiveEdgeInduction is the shared induction principle. From every prescribed implementation base, construct a legal related source base. For every construction-reachable related origin and every actual implementation successor, extend that exact source prefix legally, retain its previous choices and identities, restore the relation, and establish equal extended canonical histories. Induction over actual finite edge ancestry gives origin coverage. No fair extension or future cycle is assumed; deadlocked and terminal finite prefixes remain in the domain.

Core.ConstructiveOriginCarrier fixes actual origin occurrences and their finite edge ancestry independently of correspondence success. A non-identity designation requires an occurrence-level transport proof preserving the represented state, roots and choices and covering every actual continuation through the declared continuation or commuting-step correspondence. Equal histories alone do not establish that transport, and no future input, edge or fairness may be assumed to obtain it.

For the registered message profile, J.LocalOriginContracts C0–C5 discharge that induction through the following evidence.

Contract	Information that crosses the boundary
C0 Base and environment	Legal matching bases and a source counterpart for each actual environment step under the fixed choices
C1 Roots and attribution	Complete requests and absences, operands, dynamic owners, NB attempts and correct frontier activation
C2 Settling and selection	A legal matched snapshot and a complete coherent inventory of selected actions and outcomes
C3 Boundary realization	Primitive legality, resource residues, atomic effects and frames that compose into one legal source segment

Contract	Information that crosses the boundary
C4 Update and preparation	Simultaneous registered update and finite legal owned preparation restoring the relation
C5 Canonical translation	Complete group matching and both directions of normative generator/path translation

C3 is where separate local legality becomes joint executability. A finite acyclic descriptor graph orders the construction. Each component describes its residue after any predecessor-closed subset of completed descriptors. For a minimal remaining descriptor, its participants establish the actual source primitive's premises and exact effects; unaffected components are framed. Shared writes need one joint owner or a proved combined update. This prevents mutually inconsistent local witnesses from being treated as a global execution.

C4 restores the relation using the actual source control and return rules. Prepared local words compose because their writes are owned and their operands do not depend on another process's newly prepared requests. A terminal export may end at its completed L4 checkpoint without inventing a next L1 window. If a real later successor exists, its case must still establish the legal phase entry it consumes.

C5 proves equality of the semantic histories, not just equal event lists. It preserves whole groups, exact identities and values, old history restrictions and all required paths in both directions. J.LocalBoundaryCompose, J.LocalOriginCanonExt and the update/preparation proof together establish J.LG-Origin. They supply no progress conclusion.

From origins to all finite prefixes

Core.RegisteredMessageBatch is an explicit implementation completion profile, not a consequence of calling a design synchronous. All selected completions, NB retirements/results and the simultaneous update at an edge become complete together. Between edges, silent activity does not change the selected canonical history. The initial state and every actual completed edge, including a terminal edge, have a qualified origin.

J.BatchPrefixCoverage therefore associates an arbitrary finite prefix with the initial history or its last already completed edge. It does not finish the current cycle, require future input, or select a fair continuation. Combining this lift with J.LG-Origin gives exact all-finite-prefix history coverage for the declared Batch domain. Co-completion does not merge independent groups into one atom or add order between them.

A monitor using this guarantee assembles the complete edge bucket before evaluating a transferred property. Its result must still be extensional on the canonical history: changing erased timing or incidental listing order cannot change the verdict. A cycle-count bound is not made extensional by batching. A finer implementation carrier or monitor exposing partial inventories needs its own exact correspondence proof.

The retained alternative J.ConstructiveOriginIdealSafe uses cycle-origin coverage, the appropriate safety carrier, Core.RTLPartialCycleClosure and J.CycleIdealCover. It additionally needs J.IdealPrefixClosed and the stated origin inclusion/transport conditions. Exact Batch coverage does not require that property closure. The two lifts should not be conflated, and neither may silently narrow an already declared implementation domain.

Locality theorems and their limits

J.OrdinaryConstructiveComposition provides a stronger organizational result for the prospectively defined OM and OA profiles. It assembles the global relation from process-local LP0–LP5 evidence, ordinary primitive refinements, G1–G7 integration facts, settling/matching and environment/carrier evidence. Within that scope,

no extra application-specific global correspondence invariant is an input. Local evidence is substantive and universally quantified over the admitted incident interfaces; it cannot hide a peer's private state, future response or the desired global continuation.

OM has independent ordinary FIFO/rendezvous resources and message-only observation. OA adds supported unary waits, paired synchronization and qualified sequential signals with pre-update sampling. These locality profiles exclude shared capacity and endpoint ownership, shared mutable application memory, dynamic reset, general configuration/R2C extensions, unsupported forwarding/resolution and selected START/FINISH conventions. Such features are not declared universally illegal; they require a broader qualified construction.

J.ElaboratedConstructiveComposition gives ELM/ELA locality for separately owned FIFOs and the specifically qualified MoveAB, request-fed JoinFire and FIFO-fed ForkABC schemas, while retaining the other ordinary restrictions. It requires Q.SafetyQualified, expanded local packet/bookkeeping evidence, complete NB decision binding and a closed generator inventory. It is not a theorem for arbitrary elaboration, shared pools or zero-storage compounds. An internal move with no process completion still needs its register-update and local-preparation proof.

These are conditional paper-level composition results. The rules do not assert that every R/E-correct implementation admits the required local evidence, or that a concrete HLS flow, automatic extractor or mechanization has already been qualified.

5 Selecting observations and feature extensions

The general constructive architecture has named extensions beyond the ordinary locality theorems. Select the actual feature and observation profile before proving coverage; do not infer a combined result by conjoining independently proved relations with incompatible roots or state.

Selected feature	Controlling interface or theorem	Essential qualification
Registered messages	Core.RegisteredMessageBatch; J.LG-Batch-Safe	C0–C5 and actual whole-edge completion semantics
Stable direct inputs with message observation	J.StableDirectInputContracts; J.StableRead-Batch	Real initialization anchors, stable values, in-place substitution and legal source lifting
Waits, paired synchronization and signals	Core.RegisteredAnchorBatch; J.LG-AnchorBatch	Unary waits and pairwise SyncChannel/ac_sync only; A0–A6, complete anchor membership and joint update
Direct inputs with retained Reads	J.AnchorDirectInputLift	Anchor-determined Read descriptors retained in whole groups; per-interval value binding and old-work isolation
Transformed internal input payloads	Core.LogicalPopPayload; Q.CT.LogicalPayload and LogicalPair; J.LogicalPayloadRefinement	Causal ownership, physical representation and legal later source use; safety-only use retains the representation and operational clauses

Selected feature	Controlling interface or theorem	Essential qualification
Configuration records	Core.AnchorConfigObservation; J.LG-ConfigBatch	CO and CVG contracts for each governing record and every actual use
Dynamic edge-sampled reset	Core.RegisteredResetBatch; J.LG-ResetBatch	RB0–RB5, scoped disposition, epoch identities, hold/release and initialization
Edge-observed combinational signals	Core.RegisteredR2CBatch; J.LG-R2CBatch	RC0–RC6, joint acyclic settling and exact emission/participation

For ordinary retained sequential Reads, RULE 1 fixes membership and value at the preceding anchor; a later message-dependent branch cannot retroactively add a Read to a completed anchor group. Multiple Writes sharing a succeeding anchor are folded in executed source order, with the final Write supplying the retained value. Physical samples do not create extra logical Reads or move their anchors.

J.AnchorDirectInputLift supplies the retained-Read option when direct inputs coexist with observed anchors. It substitutes the legally governing value in place within each interval while keeping the complete anchor-determined Read descriptor in the synchronization group. Every physical sample and old-work operand must meet the stability/update-isolation contract. Conditional or repeated evaluation is allowed only with the qualified anchor-determined membership; value equality alone cannot establish that membership or justify deleting a Read from a retained group.

StableRead handles a narrower message-only case by substituting a fixed configuration value at each reached read expression and lifting the reduced proof back to the original source. It does not hoist conditional evaluation or waive Core.16. ConfigBatch instead records designated configuration values at real initial anchors or governing paired transactions. Designated uses create no observable process-Read labels in that option. Every actual use must still receive its governing value, and old work must be isolated from newly presented pins. Zero-length intervals and conditionally unused fields still have their declared configuration records.

Reset matching preserves actual scope, priority, cancellation and storage effects. Discarded contents are not fabricated Pops; balances restart only for affected channels. Partial reset across live communication requires a modeled cross-epoch disposition. History coverage does not require eventual reset release. Asynchronous assertion needs its separate implementation argument, and safety across reset does not prove preservation of outstanding K workloads.

R2C observation is of the qualified fixed point, not transient delta-cycle propagation. Ordinary emission uses participation plus first observation, changed value or a fresh sampler; cycle-locked emission requires a unit in every participating cycle. The reset ledger restarts the appropriate comparison state without renumbering prior history. A repeated numerical value does not excuse an omitted required unit. The combined constructive profile requires compatible local relations, inventories and the complete acyclic settling graph.

For transformed internal input paths, Core.LogicalPopPayload and the representation and operational clauses of Q.CT.LogicalPayload permit causal original-value attribution at the actual logical Pop retirement without requiring raw-value recovery from current registers. J.LogicalPayloadRefinement supplies the corresponding local safety cases without completion obligations. The proof still needs actual admission/ownership, physical representation, legal later source use and all remaining constructive cases. Physically sampled observations are excluded; projection cannot relabel them. Q.CT.LogicalPair and, where selected, Q.CT.PairedRepresentation

preserve both logical owners and one complete atomic return packet for a computed pair. Reusing a property proved on an ungrouped source requires a separate group-erasure/inclusion proof covering observations, required order and admissible cuts; simultaneous selectability of two Pops does not supply that proof.

The current shared-memory safety route needs J.Mem lowering to explicit value-labeled channel or anchored-signal operations. A general memory-state extension is discussed but is not supplied by the current theorem; using one would require a separate proof. Synchronization alone establishes neither read-value equality nor that lowering. Transaction-changing mappings additionally use Core.MemProfile/Core.MemFaithful. Core.MemSyncOrder remains a separate scheduling-conformance obligation; it does not add raw memory events to the constructive alphabet. Completion must expose hidden memory backpressure and service conditions. Multiple clocks remain outside the single-clock theorem instance; Appendix O is an informative sketch, not a CDC composition theorem.

6 Source properties, testbenches and stimulus classes

What the source verification must establish

J.ReachableHistCov uses the fixed outer history CanonHist_Σ_DUT: every implementation prefix κ in the declared domain D has a legal reachable source prefix λ with equal CanonHist_Σ_DUT histories. J.SafeTransfer-HistCov transfers an extensional predicate Φ once Φ is independently established on every reachable source history. The generic-H interface is J.ReachableHistCovOn(D,A): H and the source domain A are independently bound, and every κ in D has λ in A with H(κ)=H(λ). J.SafeTransfer-HistCovOn supplies its property transfer. The correspondence proof does not establish functional correctness of the source design.

Core.SourceCheckpointProfile can narrow the source-property domain to independently declared initialization or completed-cycle checkpoints ChkRef. J.SourceCheckpointRepresentative proves that constructed witnesses have equal-history representatives there; the actual constructed prefix remains the inductive state. J.SafeTransfer-HistCovOn then transfers a property proved throughout that domain. Ordinary cycle-granular simulation alone does not prove the representative theorem or cover all unresolved source choices.

A second property-discharge option uses one selected complete source execution ρ^o. It needs J.RefProjCover, or RefProjCoverOn for the declared source domain: every relevant source history must be a finite ideal of $H(\rho^o)$. It also needs Φ on every finite canonical ideal of $H(\rho^o)$. Checking ordinary prefixes of that one run is insufficient. A directly proved all-ideals premise requires no ideal-closure assumption; deriving it from one successful finite-history evaluation does require the separate closure argument. Restricting the covering domain does not weaken the all-ideals property premise of this transfer clause.

For example, a scoreboard can evaluate an occurrence-keyed request/response ledger after merging a complete edge bucket. It must express matching and values from H and be invariant under every timing/order distinction that H erases. If a needed relation cannot be recovered from H, select and qualify a richer observation or use another property theorem. A logger cannot invent a canonical dependency to make the property transferable.

Reusable stimulus-class correspondence

J.ProcessEnvClass fixes a family of stimuli, initial data and choice resolutions. The primary profile includes the testbench processes in both compared systems; the implementation-side testbench need not be synthesized, but its runtime, libraries, wrappers and interface semantics must be qualified. Reactive responses follow from related testbench state and actual return values. Identical testbench source code alone does not establish that correspondence.

An external-testbench profile is also available when its explicit environment relation supplies legal source counterparts for every actual environment step and consumed observation. It is not automatically a cheaper or weaker contract. Neither placement permits substitution of a more cooperative environment or reclassification of reactive responses as freely reselectable stimulus.

J.UniformConstructiveEvidence organizes class-wide obligations: UCL0 binds the family, instance semantics, history and source-property domain; UCL1 supplies uniform local construction over every actual successor; UCL2 establishes environment/testbench composition; UCL3 supplies complete prefix and projection coverage. Class membership must be independent of constructor success. The named locality theorems can discharge their portion of these obligations within scope.

J.UniformConstructiveCoverage then gives history coverage for every admitted stimulus and finite prefix. J.ConstructiveClassSafe adds the independently proved source property throughout the class. Applying that result to a member needs exact binding and residual membership/applicability checks, rather than a fresh mathematical correspondence proof. Sample campaigns, a reusable structural artifact or one testbench program do not establish these universal premises. A completion-class claim separately binds and proves its workload, fact, agreement and progress obligations.

7 The bridge from safety to completion

Completed facts rather than outputs alone

K consumes J.InternalFactCorr over an explicit signature F. F includes R's acceptance facts, every workload unit's member completions, and every historical control/value predecessor used to identify those operations, recursively. It includes failed NB outcomes or other non-transfer observations whenever the completion proof uses them. Initial facts are bound separately.

J.ConstructiveFactEvidence adds CF0–CF3 to the selected constructive route: matching base facts and stable identities; an exhaustive actual-edge inventory; legal local source realization and preservation of all old facts; and an exact finite-prefix lift for the entire F ledger. Core.RegisteredFactBatch and Core.RegisteredFactPrefix provide that lift in the qualified registered specialization. Outer message Batch coverage alone does not prove that all internal results have the same prefix semantics.

J.ConstructiveFactCorr derives J.InternalFactCorr; J.RegisteredConstructiveFactCorr specializes it for qualified message/anchor constructions. Other extensions need their own full fact cases. Fact correspondence establishes what completed, with what identity, value and order. It does not establish current request presence, microstate equality, enablement or fairness. Those remain explicit conformance invariants.

Agreement is conditional on occurrence

Core.FinitePrefixProvenance gives a common fact-key universe, legally fixed roots, deterministic typed evaluators and a well-founded predecessor relation. Its independently declared domain includes both the workload witness and all source prefixes supplied by internal-fact correspondence. Occurring facts must be available with their historical values, and all evaluator predecessors must be legally available; future results and retroactively reselected observations are forbidden.

Core.FinitePrefixAgreement proves that the same available fact has the same value in two such prefixes. J.InternalFactAgreement transports that conclusion to matching RTL facts. This says nothing about whether a missing operation ever occurs. An evaluator for an operation also determines whether its descriptor exists, including its control/iteration decision; agreement is not merely payload equality after assuming the desired operation exists.

The default serial workflow consequently needs common timing-independent provenance for consumed control and values, or a separately proved observation/outcome alignment with equivalent coverage. An NB arbitration winner or signal sample that varies with issue timing can invalidate that agreement even if each execution is individually legal and safe. A deterministic simulator, B1, or one accepting source run does not repair the missing premise.

The logical-payload safety profile described in §5 can also supply facts to this completion bridge, but only with the complete CF0–CF3 and applicable fact-carrier cases. Its logical values are fixed causally before comparison with the workload witness. Representation alone supplies neither agreement nor a legal witness lift, exposure, delivery or service; all separately triggered K.CT obligations remain.

8 Required-workload completion

The finite witness and prerequisite graph

K.CT.Instance fixes finite required testbench acceptances R prospectively, with their logical keys, values and per-endpoint order. Acceptance is an observed receiving event, not merely output assertion or a DUT-side write. The clock, local updates and required environment service continue until acceptance. No affecting reset, cancellation or premature termination may remove an outstanding required obligation without a separate preservation qualification.

K.CT.Witness supplies a finite legal execution of the same Sys_ref accepting R and a finite set W of commit units containing R and its prerequisites. Every W member actually occurs in that witness. Units and finite predecessor sets are fixed before the implementation execution being proved, form an acyclic graph, and retain all consumed resource/control dependencies or their separately proved invariants. Completed obligations remain completed in the bound epoch.

The default K.CT.SerialWitness uses Core.PolicyS: within one synchronization interval, an owner does not issue a later blocking Push/Pop until every earlier source-order blocking Push/Pop in that interval has committed in a strictly earlier cycle. Independent owners remain concurrent. The source model uses the bound capacities and real enabling laws, or a separately proved smaller-capacity lift. The witness stops at a finite prefix satisfying Core.RequiredAcceptanceObserved; it need not drain every FIFO, finish trailing loop-head Pops or satisfy a maximal-run response monitor.

A primitive rendezvous is one unit containing both endpoints. Genuine atomic groups remain indivisible. When same-owner atomic members cannot be issued under ordinary Policy S, K.CT.GroupSerialWitness uses a qualified group-serial restriction of Policy L: all independently prepared members issue legally in one pre-settle window, while every preceding group required by each participating owner’s source order has committed in a strictly earlier cycle. Groups cannot cross synchronization or reset boundaries. This is not a redefinition of Policy S and does not remove unresolved sibling dependencies. Qualified overlapped witnesses retain their own preparation and residual obligations.

K.CT.Closure is stronger than collecting dependencies visible in one favorable schedule. For the standard profile it includes lower-ordinal endpoint operations; source-earlier blocking calls on a sequential owner’s path; all required I/O in preceding stages and older iterations of a pipelined owner; matching inputs; guaranteed-floor space releases; and all consumed control/value and sync/gate predecessors. Partial-stage sibling omissions need the explicit legal pre-activation preparation qualification. These dependencies organize a sufficient proof; they are not commands forcing RTL to execute serially.

Exposure, service and the theorem

K.CT.Exposure says that, for any incomplete W unit whose predecessors are complete, every admitted service-qualified continuation either completes it after finitely many steps or reaches continuous actual enabling until it completes. Atomic and rendezvous units require simultaneous whole-unit enabling. Persistent requests alone are insufficient if resources or guards keep disappearing.

K.CT.Service supplies service of every continuously enabled required unit, together with the separately declared internal service for preparation, computation, capture, movement, entry and restoration. Fairness for separate endpoints does not establish fairness of an indivisible transaction. Repeated but intermittent readiness is not continuous enabling. The proof must establish actual concrete opportunities as well as the service assumption selecting them.

K.CT.Completion is a finite topological induction over W. Once predecessors complete, Exposure and Service complete the next unit. Facts and Agreement identify its occurrence, value and endpoint order. Thus all required acceptances occur. Service may be required only through R; completing additional W members beyond that acceptance point requires the stronger service scope stated in K. The theorem gives no uniform time bound.

K.CT.Standard discharges Exposure for the declared B.OF/B.CT/Q.CT profile. It first uses K.CT.Alignment: agreed historical facts determine the source-required operation and descriptor, and B.OF.Control/B.CT.Control bind that decision to actual finite legal controller preparation, with B.OF.Control supplying the source-aligned pipeline membership and iteration rules. PlacementLemma then derives physical placement and eventual permanent isolation after relevant older work exits. Request, resource, residual/joint and synchronization lemmas establish persistent enabling. The controller cannot make an obligation vacuous by simply omitting the source-required operation.

Why guaranteed capacity matters

B.CT.FIFO distinguishes nominal modeled storage from guaranteed usable space. With an exclusive ingress queue of F entries and optional outstanding credit C, a positive guaranteed floor g satisfies $g \le F$ and, when applicable, $g \le C$. Reserved slots and downstream pipeline storage do not automatically enlarge this floor.

For a standard initially empty positive resource, W includes Push(k) before Pop(k), and Pop(k−g) before Push(k) when $k > g$. Those space edges apply even when the witness used the full nominal reference capacity B. After adding them, the graph must still be acyclic. For example, nominal capacity four with a justified floor two requires Pop(1) before Push(3); a source pass relying on all four nominal slots does not waive that prerequisite.

Running the witness at a smaller capacity is optional and additionally requires K.CT.Capacity's legal lift into the fixed reference, supported on both source and implementation sides. The safety model keeps the actual staged structure. An unsuccessful conservative witness or cyclic sufficient prerequisite graph is inconclusive, not a deadlock verdict. B=0 uses joint rendezvous semantics, never substitution into positive FIFO floor arithmetic.

9 Qualifying pipeline and resource progress

B.OF applies to the declared ordered finite-stage no-wait pipeline profile. It preserves iteration order and exact retirement state, requires serviced finite local computation, and ensures completed downstream work can drain independently of blocked upstream work or new input. Bubble compression is permitted but not required: an empty location behind incomplete downstream work does not by itself force advance. An unconditional global stall that indefinitely freezes required completed downstream work fails this profile.

B.CT adds source-aligned controller behavior, exact storage ownership and endpoint order, capture and guard laws, atomic compatibility, synchronous update semantics, synchronization/restoration and service. Its laws apply over all admitted reachable states and surrounding activity, including activity outside W. A no-wait body, synthesis directive or B-faithfulness alone does not establish them.

Resource organization	Named progress qualification	Distinction to preserve
Queue with capture at the Pop site	B.CT.FIFO and B.CT.Capture	Split capture retains the item; coincident retirement uses the old head
Independent early input paths	Q.CT.InputStaging and InputStagingProgress	Physical forwarding precedes logical retirement and must deliver an isolated target
Equal-length coupled positive pair	Q.CT.CoupledInputStaging and K.CT.CoupledDeliveryLemma	Paired movement and retained arrivals require joint internal service
Fixed-offset late second read	Q.CT.StaggeredInputStaging and K.CT.StaggeredDeliveryLemma	Track the early-only prefix, matched late read and paired suffix separately
Synchronization-gated regions	B.CT.Sync and Restore; Q.CT.Epoch; K.CT.SyncLemma	Restoration cannot depend circularly on traffic through the closed gate
Observation-neutral logical retry	Q.CT.Retry	Attempts implement one qualified persistent obligation under its own resource scope

B.BoundaryBinding distinguishes physical admission/movement, process-facing Issue and retirement, and legal source launch. Accepted unmatched input storage belongs in the channel accounting even when physically inside a pipeline. Registers after the selected Pop are process state. The last source-ordered message Issue of iteration j cannot follow the first message Issue of iteration j+1. For a regular unstalled schedule with fixed issue offsets, last_offset − first_offset ≤ II. This constrains source-facing Issue, not physical stage numbers; stalls, delayed assertion, legal groups and the selected boundary map still require the actual E3 proof. Deep physical read offsets do not excuse that constraint; a qualified early-storage interpretation may be needed.

B.CT.Guard allows immediate enabling under persistent support, or a finite-response realization with an actual internally serviced, non-replenishable decreasing measure. Joint units need the joint guard proof. Once true, the guard remains true while its support and the request persist, until retirement. Persistent support is a separate conclusion of the applicable ResourceLemma, InputStagingProgress or delivery proof; the guard law does not establish it. A generic assertion that the transfer will eventually happen would merely assume the result the exposure proof needs.

Sync/epoch qualification preserves resident items, credits and global matching order. A fully drained boundary is one sufficient case, not a universal requirement. Gate restoration must be noncircular, and the protected target cannot lose availability to a later withdrawal even before its request becomes pending. ClockRefinement is an optional exact interleaving presentation for its narrower canonical capture/guard/advance profile; matching finite edges alone does not transport progress.

The named input-path alternatives are not freely interchangeable. Computed representation does not supply delivery; coupled paths need joint movement rather than separate per-input fairness. Fixed-offset staging does not qualify variable offsets, arbitrary unequal paths, staggered process-facing retirements or unconditional global stall. Mixed positive/B(0), retry and epoch-staging combinations retain their explicit compatibility requirements. Completion service establishes no sustained initiation interval or throughput rate.

K.CT's "Explicit exclusions and extensions" gives the standard-profile list: "Standalone non-retry NB attempts as W units, multi-member non-atomic residual units, early capture outside the applicable Q.CT.InputStaging, Q.CT.CoupledInputStaging or Q.CT.StaggeredInputStaging clauses (whether single- or multi-member), recursive readiness networks, shared credit pools, general arbitration, timing-dependent NB/signal control, resets crossing outstanding obligations, and synchronization among more than two participants require additional stated laws and proofs." These exclusions do not exclude otherwise qualified NB observations from safety; a consumed NB predecessor still needs its own completion discharge. Q.CT.Extension is the explicit route for those additional laws and proofs. Failure to fit the standard profile does not prove the design incorrect, and safety eligibility of a feature does not establish its completion eligibility.

10 Snooping and latency-sensitive choices

Appendix L remains relevant when hidden outcomes affect later visible behavior or attribution. L.WC records witness-compatible ε choices, including failed-call identities used by E3/E5. NB-CF, snooping, direct construction and a complete compatible per-site mixture are ways to establish the relevant facts. The direct constructive route derives outcomes and call attribution from local contracts and matched snapshots; it does not automatically require an additional Appendix L package.

A mixed per-site discharge needs a complete partition within one bound instance and one joint construction. Selected outcomes must be simultaneously legal under the common settled snapshot and source decision law, with compatible choices, footprints, dependencies and atomicity. Independently proved per-site outcomes do not establish that joint legality.

L.JointOptionality is a separate set-level proof for omitting designated snoops: all admitted joint variations must preserve the selected canonical DUT history at independently matched logical cuts, with separate legal witness/cut availability and coverage. Optionality at each site while all other sites remain snooped is insufficient. A class claim requires the proof uniformly over the class. Boundary-only optionality does not erase wider proof-internal or internal-testbench matching obligations, and it waives no K premise.

Snooping must select a legal source behavior. It cannot override Core.NBDecision, change fixed stimulus, suppress a mandatory success, substitute payloads or bypass source phase restrictions. Cadence-sensitive retry counts, timeouts or arbitration need the qualified latency-sensitive protocol/alignment treatment; calling failures ε does not erase their effects on later behavior.

When a snooping harness is used, W1–W5 require recorded per-ordinal observations, source-side mirror gating, qualified registered sampling, exact identity/kind/payload checks, outstanding-event accounting and complete choice coverage. L.Dir keeps the RTL free-running and requires ordered source consumption of its recorded observations. A live AND of current signals can lose or alias events when the source lags. An unproved watchdog bound expiring is an inconclusive witness-realization failure, not a proved design deadlock. A finite clean log does not prove future consumption or universal correspondence.

K has no standalone snooping premise. It consumes the selected J adapter with all that adapter's applicable premises, plus K.CT.Agreement and progress. Aligning one source witness with one observed RTL run does not prove agreement between a serial workload witness and every RTL-matched source prefix. A separately

proved alignment may establish the required agreement, but recording outcomes alone does not do so. Snooping therefore neither replaces K's agreement nor proves exposure and service.

11 Applying and reviewing the framework

Start by declaring the intended claim: a safety predicate and observation, a required acceptance workload, or both. Fix the source/implementation artifacts, environment placement, choices, clock/reset, capacities and elaboration. Choose an actual completion carrier and feature profile. A narrowed observation can simplify a proof only when the property remains expressible and whole-group/order obligations are preserved.

For safety, establish source well-formedness and boundary/memory realization, then local process and primitive evidence, integration, operational settling/matching, environment and carrier coverage. Use ordinary or elaborated locality where its restrictions fit; otherwise discharge the broader named constructive contracts. Independently verify the actual configured source property on Reach_ref or qualified ChkRef, or establish the covering-execution alternative with its full premises. Bind monitor behavior to that same history semantics.

For completion, add the full fact signature and exact fact-prefix correspondence; provenance/agreement including the workload witness; finite acceptance witness and complete acyclic closure; source-aligned controller and preparation evidence under B.OF.Control/B.CT.Control; actual resource floors, delivery, guards and atomic/sync compatibility; and boundary/internal/environment service. A source pass is only one input to this chain. Optional drained-test diagnostics do not replace acceptance or prove progress: empty monitored FIFOs alone do not establish a complete empty-inventory observation.

M.7a is the authoritative side-condition ledger. Each consumed premise has an owner, evidence, validity domain, trigger and invalidation rule, or a proof that its trigger is false. Framework-derived lemmas are not extra assumptions. One reusable artifact can discharge several named owners, but their obligations remain distinct. Tool/runtime qualification binds relevant versions, libraries, scheduling and reset/X semantics; unverified assurances belong in the trusted base.

Evidence level	Typical responsibility
Framework	Prove generic construction, projection, agreement, composition and workload-induction lemmas
Tool or component family	Prove qualified transformations, primitives, controller laws and runtime semantics over their declared domains
Design or stimulus class	Supply local process invariants, integration/environment evidence and independent source-property proofs
Concrete application	Bind generated artifacts, topology, parameters, boundaries, class membership, workload and remaining instance-specific evidence

Core.EvidenceRealization distinguishes a mathematical proof family, a sound total extractor on a qualified domain, and individually checked outputs of a partial generator. A partial generator's failures cannot be excluded from a universal implementation claim merely because no certificate was produced. Core.ExtensionConsumerCoverage requires every admitted extension to have each applicable consumer case or a proved inapplicability; a safety extension with no fact or progress case is not automatically completion-qualified.

A useful review follows the dependencies rather than counting checks. Ask whether source legality is independent of construction success; whether every actual successor is covered; whether all consumed observations and facts are present; whether property validity is separate from transformation correctness; and whether each progress step has both a real opportunity and service. Reuse remains valid only while its semantic and artifact bindings remain within the proved domain. The current rules specify these proof interfaces; they do not claim a completed universal tool qualification or an unconditional verification-cost advantage.

12 Where to read next

Question	Current rules location
Which user design constraints matter	Appendix F and the main scheduling rules
What models and source steps are legal	Normative Formal Core; Appendices B, C and G
How local evidence produces global safety	J.LocalOriginContracts; J.ConstructiveEdgeInduction; J.LG-Origin; J.BatchPrefixCoverage
When composition is genuinely local	J.OrdinaryConstructiveComposition and J.ElaboratedConstructiveComposition
How source safety and class coverage are discharged	J.SafeTransfer-HistCovOn; source-property monitor contract; J.ProcessEnvClass; UCL0–UCL3
How internal facts connect to workload completion	J.2c; Core.FinitePrefixProvenance; K.CT.Instance through K.CT.Completion
How the standard exposure proof works	B.OF.Control/B.CT.Control; B.OF and B.CT progress laws; K.CT.Alignment and Standard; Q.CT; Appendix N
How choices and snooping are qualified	Appendix L and the L.WC/L.Dir entries in M.7a
How elaborated boundaries and accounting are bound	Core.ElabProj and Appendix Q.1
What evidence and mechanization remain	M.7/M.7a; Appendices R, S and T

The controlling dependency is consistent throughout: legal local source realization establishes correspondence; an independent source proof establishes safety; full facts and agreement identify the workload’s required operations; concrete exposure and service make those operations complete. Each application must preserve those boundaries while instantiating the parts it actually consumes.